

Beyond Text: Visual Description Assembly by Probabilistic Model for CLIP-based Weakly Supervised Semantic Segmentation

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Abstract

Contrastive Language-Image Pre-training (CLIP) offers a new paradigm for Weakly Supervised Semantic Segmentation (WSSS) by generating Class Activation Maps (CAMs) from text-image alignment. Existing methods primarily rely on hand-crafted templates or general attribute descriptions generated by a large language model to construct text prototypes for querying visual features. However, these strategies faces two major limitations: the inherent modality gap in CLIP prevents text prototypes achieving tight alignment with visual features; and their static text prototypes cannot adaptively respond to target instances that exhibit diverse visual attributes. To address these challenges, our key insight is to directly construct instance-specific visual description prototype as query, thereby bypassing the sub-optimal static text description optimization. To this end, we propose the Visual Description Assembly (VDA) framework. It employs a probabilistic model to map complex CLIP visual features into a structured latent space. This latent space allows us to explicitly disentangle and aggregate varied visual attributes, and then dynamically assemble them into instance-specific visual prototypes. Furthermore, to enhance the robustness of this prototype, we adaptively incorporate the semantically stable text prototype into it as the final query for generating superior CAMs. Experimental results show our method outperforms existing baselines, achieving state-of-the-art performance on WSSS benchmarks. Code is available at [here](#).

1. Introduction

Weakly Supervised Semantic Segmentation (WSSS) effectively reduces the high cost of pixel annotations by leveraging image-level labels [2, 12, 20, 29, 44]. The mainstream WSSS paradigm generates Class Activation Maps (CAMs) from classification networks, refining them into pseudo-labels for segmentation training [24, 54]. However, limited by the insufficient semantic information from image-level

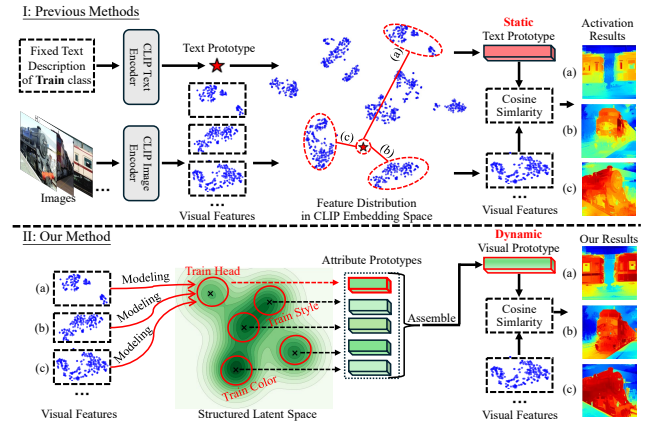


Figure 1. (I). Previous methods use a single static text prototype as query to activate visual features, but it often lies outside the visual feature clusters, leading to incomplete CAMs. Additionally, visual clusters contain similar attribute (e.g., ‘the head of train’, marked by red dashed circle) are scattered in the CLIP space, resulting in inconsistent activation results. (II). Motivated by the idea that a good query should be derived directly from the visual space with target-specific attributes, our method models raw visual features into a structured latent space to explicitly disentangle attributes and assemble them into visual description prototypes as query.

labels, these CAMs often activate only the most discriminative object regions, failing to cover their full extent.

Recently, Contrastive Language Image Pre-training (CLIP) [25] has revolutionized many advanced tasks [13, 21, 30, 31]. In WSSS, CLIP leverages the vast pre-trained knowledge to construct the text-image relationship and achieve better CAMs beyond traditional classification networks [17, 18, 37, 42, 48]. These CLIP-based WSSS methods use CLIP text feature as query to activate CLIP visual features. For instance, CLIP-ES [17] and WeCLIP [48] typically rely on simplistic, hand-crafted templates (e.g., ‘a clean origami of [CLASS]’). But such templates only confirm object existence, lacking the specific attribute details necessary for fine-grained localization. Recognizing this limitation, subsequent work, like ExCEL [42], attempts to refine the text by using Large Language Models (LLMs) to

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generate a set of richer attribute descriptions. These richer attribute descriptions are then clustered to generate a single, more descriptive text prototype. However, it only focuses on improving the textual description, while failing to address a more fundamental flaw: *the inherent modality gap in CLIP*, where image and text embeddings reside in different manifolds [5, 28].

This *modality gap* introduces two critical issues. On the one hand, as visualized in Fig. 1(I), the text prototype (even highly descriptive one) often lie outside the actual visual feature clusters. This is because CLIP text embeddings were optimized for general semantic representation to facilitate global alignment, rather than fine-grained visual understanding, such as segmentation. This misalignment fundamentally hinders the ability of query to respond accurately to fine-grained visual details, leading to incomplete CAMs. On the other hand, these text refinement approaches are inherently static; a single, fixed general prototype, regardless of its richness, cannot adapt to the vast diversity of visual attributes across different images. For instance, as illustrated by the red dashed circles in Fig. 1(I), visual features corresponding to a similar class attribute (e.g., ‘head of train’) but from different images form distinct, scattered clusters in the CLIP space. In contrast, the static text prototype is a single fixed position that lies between these clusters. Consequently, this varying proximity to the visual clusters results in inconsistent activation strengths, as the static query cannot dynamically align well with all of them.

To solve these issues, we propose a novel framework called Visual Description Assembly (VDA) which directly construct the query prototype containing target-specific attribute descriptions from the visual feature clusters, thus bridging the modality gap. The primary challenge for achieving VDA is that the attribute-related visual features in the raw CLIP space are scattered and entangled. As illustrated in Fig. 1(I), although the visual features corresponding to the ‘head of train’ attribute appear in clusters (a), (b), and (c), they are scattered across the space rather than forming a single compact group. Moreover, within each cluster, the features of target attribute are inextricably mixed with other attributes, making it difficult to separate them. We address these challenges by mapping the complex distribution of CLIP visual features onto a structured latent space, i.e., Gaussian mixture model (GMM). This structured space allows class-varied visual attributes to be explicitly disentangled and aggregated. For any given target object, we can compute its dynamic response strengths to these attribute components, enabling us to dynamically assemble an object-specific visual prototype. Besides, to mitigate potential errors in the visual prototype, we adaptively fuse it with the text prototype, which serves as a semantic anchor to obtain the final robust query prototype for superior CAMs.

Collectively, the main contributions of this paper can be

summarized as follows:

- We explore the suboptimal CAMs caused by the modality gap and static text in previous CLIP-based WSSS, and propose a framework that directly assembles description from the visual space to replace text description.
- We leverage probabilistic model to map CLIP visual features into structured latent space, enabling explicit disentanglement and assembly of visual attributes into a dynamic visual prototype. We also adaptively fuse this prototype with the text template to generate robust query.
- Our method achieves new state-of-the-art WSSS performance with only image-level labels on both Pascal VOC 2012 and MS COCO2014 dataset.

2. Related Works

2.1. Weakly Supervised Semantic Segmentation

Weakly supervised semantic segmentation based on image level labels typically uses classification networks to generate CAM to provide dense supervision [19, 41, 43, 47]. However, classifier CAM can only cover a limited discriminative area. So the focus of WSSS is on how to extend CAM to the complete target object and many works have made innovative attempts. SIPE [2] utilizes image-specific class prototypes to construct the semantic relationships between image pixels to refine CAMs. MCTformer [38] learns the attention between multi-class tokens and image patch Tokens to generate the CAM. BRNF [24] models pixel features via normalizing flows [9, 45] to extract fine-grained semantic information to assist the classifier CAM generation. PSDPM [52] mines secondary discriminative pixels via foreground pixel estimation to generate a more complete CAM. Despite the progress made by these methods, constrained by the limited semantic information that image-level supervision provides to classification networks, the generated CAM still fails to accurately cover the target.

2.2. Vision-Language Pre-training

Contrastive Language Image Pre-training (CLIP) [25] is a model trained on large-scale image-text pairs, which shows strong generalization ability across various downstream tasks [7, 22, 23, 30, 32, 34, 40, 50]. MaskCLIP [53] generates pseudo-labels based on the alignment between CLIP text and image modalities to train segmentation network. CLIP-RC [49] transfers the image-level knowledge of CLIP to dense segmentation by introducing region-level cues. Some recent studies introduce CLIP into WSSS, achieving encouraging progress. CLIMS [37] incorporates additional natural language supervision from CLIP to activate more complete objects and suppress open background noise. CLIP-ES [17] extracts GradCAM from the zero-shot classification process of CLIP. WeCLIP [48] further explores using the frozen CLIP visual encoder as a backbone

for segmentation. ExCEL [42] leverages a large language model for text semantic enrichment to enhance the patch-text alignment of CLIP to generate better CAM. However, texts used by these CLIP-based WSSS methods are static across different images and cannot be dynamically adapted to instance-specific variations, leading to suboptimal results. Furthermore, the modality gap inherent in CLIP is also overlooked by these approaches, which implies that direct text-image alignment may not be the optimal solution.

3. Methodology

3.1. Preliminaries

CAM Generation. In CLIP-based WSSS, the CAMs are calculated by the cosine similarity between visual patch features and class text embeddings as in [42]. Given an image $I \in \mathbb{R}^{3 \times H \times W}$, let $V \in \mathbb{R}^{h \times w \times D}$ be its CLIP visual features and $T \in \mathbb{R}^{K \times D}$ be the text embeddings for K classes. The CAM M_k for class k is calculated as:

$$M_k = \text{Norm}(\cos \langle V, T_k^\top \rangle), \quad (1)$$

in which Norm indicates min-max normalization and $\cos \langle \cdot, \cdot \rangle$ denotes cosine similarity between the two terms within it. Clearly, with the visual embedding V held constant, the accuracy of CAM M_k relies entirely on the quality of text embedding T and its ability to achieve a fine-grained, tight alignment with V . Unfortunately, this is what is precluded by the static, inflexible text of prior methods and the intrinsic modality gap of CLIP. Our work focus on constructing a query prototype that replaces the static text prototype by directly integrating dynamic attributes from the visual space to better align V .

Framework Overview. The overall framework of our VDA is shown in Fig. 2. It assembles attributes from the visual space into visual description prototype, which serves as query to avoid the modality gap caused by solely using text prototype. To separate visual attributes from visual space, we first propose Visual Attribute Modeling and Decoupling (VAMD) which uses invertible neural network (INN) [1] to model the CLIP visual features to structured GMM where we can explicitly decoupling attributes within class. Then, to assemble these attributes into a visual prototype, we propose Visual Description Assembly and Fusion (VDAF). It dynamically assembles attributes based on the target response to class attribute components in GMM, and adaptively fuses the text prototype to generate final robust query prototype. This query prototype, containing rich, target-specific visual attribute descriptions, is used to activate the visual features for superior CAMs. Besides, to enhance the semantic consistency of decoder features, we design Decoder Semantic Enhancement (DSE) module which uses the class prototypes from GMM as semantic anchors to conduct contrastive learning in decoder.

3.2. Visual Attributes Modeling and Decoupling

Reliable Instance Prototypes Filtering. Our goal is to disentangle visual attributes within the CLIP space and assemble them into visual descriptions. Considering that object instances in different images carry diverse class attributes, we employ instance prototypes to capture such attribute information. Specifically, for a given image I with its frozen CLIP visual features V , the instance prototype $\mathcal{P}_k \in \mathbb{R}^{1 \times D}$ for class k can be calculated by masked average pooling:

$$\mathcal{P}_k = \frac{\sum_{x=1, y=1}^{h, w} P_k(x, y) * V(x, y)}{\sum_{x=1, y=1}^{h, w} P_k(x, y)}, \quad (2)$$

where P_k is the decoder prediction mask and $P_k(x, y) = \mathbb{I}[P(x, y) = k]$. $\mathbb{I}$ is the indicator function. However, these prototypes are derived based on decoder predictions, which may be unreliable and noisy in WSSS. Using all these prototypes directly would severely corrupt our probabilistic model learning. Therefore, to filter out noisy prototypes and ensure only reliable, semantically pure inputs are used for subsequent modeling, we employ a robust filtering mechanism based on the zero-shot classification ability of CLIP. We first define the class text set as $\mathcal{T}_{zs} = \mathcal{T}_{fg} \cup \mathcal{T}_{bg}$, comprising K foreground class prompts and N background class prompts. We encode all texts in $\mathcal{T}_{zs}$ to obtain the text embeddings T_{zs} . The background prompts follow the settings in [17, 48]. We then compute the zero-shot classification scores $s \in \mathbb{R}^{K+N}$ for the prototype $\mathcal{P}_k$ as follow:

$$s = \text{Softmax} \left(\frac{\mathcal{P}_k T_{zs}^\top}{\tau} \right), \quad (3)$$

where τ is the temperature parameter in CLIP [25]. Then we can collect a batch of reliable prototypes $\mathcal{B} = \{\mathcal{P}_k \mid s_k > \eta\}$ where s_k is the zero-shot classification score of class k and η is a confidence threshold. This filtering step is vital for constructing a reliable and semantically pure latent space for our attribute modeling and decoupling.

Hierarchical Gaussian Mixture Modeling. Our key insight is to replace text descriptions with an assembly of visual attributes. However, as we discussed earlier, the visual attributes are highly scattered and entangled in the original CLIP visual space. To disentangle this complex visual feature distribution into explicit attributes and aggregate them, we propose to map it onto a hierarchical Gaussian Mixture Model (H-GMM) using INN. GMM naturally assumes that a complex distribution (target class) is composed of a mixture of multiple simple GMM components (class attributes). And we choose INN for its bijective property, which allows for precise probability estimation and ensuring we can map latent attributes back to the CLIP visual space.

However, optimizing a single GMM for all classes and all their fine-grained attributes simultaneously is a complex

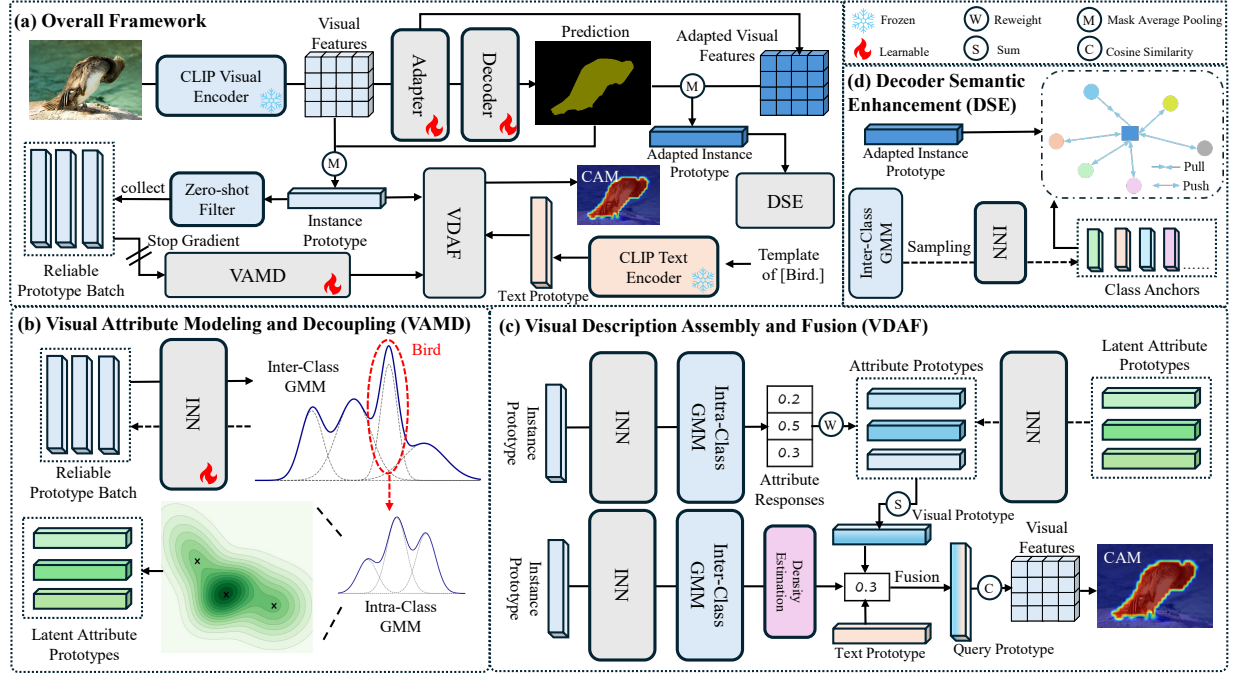


Figure 2. The overview of our framework. First, the instance prototypes are extracted from decoder predictions and collected into a reliable batch via a zero-shot filter as shown in (a). Then VAMD (b) uses an INN to map the reliable prototypes onto a hierarchical GMM (H-GMM), which models inter-class relationships and decouples intra-class attributes. Subsequently, VDAF (c) constructs an instance-specific visual prototype based on attribute responses, dynamically fuses it with the text prototype to create a query prototype, and generates a CAM for decoder supervision. Besides, DSE (d) extracts global class prototypes from the H-GMM and uses them as anchors for contrastive learning to enhance the semantic consistency of adapter embeddings.

and unstable optimization problem. Therefore, to decompose this complex task into more stable sub-problems, we adopt a progressive learning strategy. We first construct an inter-class GMM by establishing stable class centers, and only then do we build fine-grained intra-class GMMs upon this stable foundation.

We first use INN to model an inter-class GMM to establish stable class centers for the K categories in the latent space. Specifically, the INN learns a bijective mapping $f_\theta : X \rightarrow Z$, denoted as $z = f_\theta(x)$ to map the original features X to the latent space Z , and it is trained by minimizing the standard Negative Log-Likelihood (NLL) loss $\mathcal{L}_{nll}$ [9, 14]:

$$\mathcal{L}_{nll} = \mathbb{E}_{x \sim X} [-\log p_Z(f_\theta(x)) - \log |\det J|], \quad (4)$$

where $p_Z(z) = \sum_{k=1}^K \pi_k \cdot \mathcal{N}(z|\mu_k, \Sigma_k)$ means modeling the latent distribution $p_Z(z)$ as a Gaussian Mixture Model (GMM) with K components and J is the Jacobian matrix of $f_\theta(x)$. To enhance optimization stability and simplify calculations, we set all covariance matrices to be identity matrices, i.e., $\Sigma_k = \mathbb{I}$. Besides, the mixing weights π_k are parameterized as learnable parameters. This is achieved by applying the Softmax function to a learnable logit vector $\psi \in \mathbb{R}^K$, such that $\pi_k = \text{Softmax}(\psi)_k$. Substituting this

GMM prior into the NLL loss (detailed derivation in Appendix), we obtain the inter-class maximum log-likelihood loss $\mathcal{L}_{inter}$ as follows:

$$\mathcal{L}_{inter} = \mathbb{E}_{x \sim \mathcal{B}} [-\text{LSE}(c_k - E_k(f_\theta(x), \mu_k)) - \log |\det J|], \quad (5)$$

where $E_k(z, \mu_k) = \frac{1}{2} \|z - \mu_k\|_2^2$ is the negative log-likelihood of the k -th component, and $\text{LSE}(\cdot)$ denotes the logsumexp operation. The learnable parameters are the log-weights $c_k = \log(\pi_k)$ and $\mu_k = k$. Further, to associate the GMM components learned by $\mathcal{L}_{inter}$ with their corresponding classes, we introduce a class-discriminative loss:

$$\mathcal{L}_{dis} = -\mathbb{E}_{(x,k) \sim \mathcal{B}} [-\text{LSM}(c_{k'} - E_k(f_\theta(x), \mu_{k'})) - c_k], \quad (6)$$

where k' means other classes except class k and $\text{LSM}(\cdot)$ is logsoftmax operation. It pulls x to associate with its corresponding k -th component and pushes it away from other components. The optimization of inter-class GMM stabilizes the main class centers. We then build upon this stable foundation by extending each inter-class component into its own M -component intra-class GMM, i.e., $p(Z|k) = \sum_{i=1}^M \pi_i(k) \mathcal{N}(\mu_i^k, \mathbb{I})$. Each μ_i^k represents a distinct latent visual attribute for class k and $\pi_i(k)$ is the mixing weight

of i component within class k . However, learning all $M \times K$ attribute centers independently from scratch is unstable. To ensure the optimization of these attributes does not catastrophically shift the already stable class center μ_k (now denoted μ_1^k), we parameterize the other $M-1$ attribute centers as offsets $\{\Delta\mu_i^k\}$ relative to this main center:

$$\mu_i^k = \mu_1^k + \Delta\mu_i^k, \quad \text{with } \Delta\mu_1^k = 0. \quad (7)$$

This parameterization allows the optimization to focus only on learning the offsets representing attribute variations. These offset vectors are then optimized by the following intra-class loss $\mathcal{L}_{intra}$:

$$\mathcal{L}_{intra} = \mathbb{E}_{(x,k) \sim \mathcal{B}} [-\text{LSE}_i(c_i^k - E(f_\theta(x), \mu_i^k)) - \log |\det J|], \quad (8)$$

where $c_i^k = \log(\pi_i(k))$ here. Finally, we adopt a progressive training strategy for these objectives. The INN is first warmed up for several iterations using only $\mathcal{L}_{inter}$ and $\mathcal{L}_{dis}$ for modeling inter-class GMM. After this warm-up, we introduce $\mathcal{L}_{intra}$ and optimize INN using combined loss:

$$\mathcal{L}_{inn} = \mathcal{L}_{inter} + \mathcal{L}_{dis} + \mathcal{L}_{intra}. \quad (9)$$

This process yields an explicitly structured H-GMM that models stable inter-class relationships and disentangles fine-grained intra-class visual attributes.

3.3. Visual Description Assembly and Fusion

Once the INN is trained, it effectively learns a ‘visual attribute vocabulary’ for each class. The components in the intra-class GMMs act as distributions for specific latent visual attributes, which can be regarded as the core visual building blocks that constitute the class. For any given object, instead of relying on the static, one-size-fits-all text anchor T_k , we dynamically assemble a target-specific visual description by quantifying how strongly the object exhibits these learned visual attributes. This assembly process involves the following three steps:

Visual Attribute Prototype Retrieval. The learned attribute prototypes $\{\mu_i^k\}$ exist in the latent space Z . However, to be used as a query, they must ultimately be compared against the original CLIP visual features V , which reside in the visual space. Therefore, we leverage the invertibility of INN $g_\theta = f_\theta^{-1}$ to map these latent attribute centers back into the CLIP visual space:

$$a_i^k = g_\theta(\mu_i^k). \quad (10)$$

These M prototypes represent the compression of certain abstract attributes of class k in the CLIP visual space, such as color, shape, and action, etc.

Attribute Response Strength Calculation. The previous

step has retrieved the visual vocabulary $\{a_i^k\}$. However, a specific instance in an image will only exhibit a subset of these general attributes. For example, the object may strongly match the certain attribute but not the other one. Therefore, before we can assemble a description, we must first quantify how strongly the current object $\mathcal{P}_k$ responds to each of these M attributes. These response strengths will serve as the precise mixing weights, allowing us to build a composite prototype that only includes the attributes the object actually possesses.

Specifically, for the given instance prototype $\mathcal{P}_k$. To determine how it relates to the discovered attributes, we first map it to the latent space: $z_k = f_\theta(\mathcal{P}_k)$. Next, we calculate the attribute response strength by computing the posterior probability of the latent prototype z_k belonging to each of the M intra-class GMM components. This posterior probability ω_i^k can be calculated as:

$$\omega_i^k(z_k) = \text{Softmax}_i \left(-\frac{\|z_k - \mu_i^k\|_2^2}{2} + c_i^k \right). \quad (11)$$

Each $\omega_i^k(z_k)$ represents the response of the target prototype $\mathcal{P}_k$ to the i -th visual attribute within class k . With these instance-specific response strengths now quantified, we have all the necessary components to construct the final composite visual description.

Composite Visual Attribute Assembly. Now we can assemble the final composite visual description A_k^{vis} , which serves as a query prototype dynamically tailored to the specific visual manifestation of object. This composite prototype is computed as a weighted sum of all visual attribute prototypes $\{a_i^k\}$, using their corresponding attribute response strengths $\omega_i^k(z_k)$ as the weights:

$$A_k^{vis}(\mathcal{P}_k) = \sum_{i=1}^M \omega_i^k(z_k) \cdot a_i^k. \quad (12)$$

The resulting vector $A_k^{vis}(\mathcal{P}_k)$ is a dynamic, instance-specific visual description, assembled purely from the learned visual attribute space.

Density-Based Adaptive Prototype Fusion. Ideally, the high quality A_k^{vis} would always be superior than text. However, we must confront a core challenge of WSSS: the quality of A_k^{vis} is entirely contingent on the quality of the input prototype $\mathcal{P}_k$, which is derived from potentially noisy prediction masks. If $\mathcal{P}_k$ is incomplete or erroneous, A_k^{vis} will be a flawed, unreliable description. Although the template text ‘a clean origami [CLASS]’ can only provide a general semantic representation, it is semantically stable and can serve as a semantic anchor. Therefore, we believe that the reliable T_k should not be simply discarded but be dynamically fused to the visual prototype.

Specifically, we define a adaptive, density-based weight $\alpha_k(\mathcal{P}_k)$. This weight measures how ‘typical’ the latent prototype $z_k = f_\theta(\mathcal{P}_k)$ is under its corresponding inter-class

component, $\mathcal{N}(\mu_k, \mathbb{I})$. We compute this by taking the probability density of z_k and normalizing it by the maximum probability density (when $z_k = \mu_k$):

$$\alpha_k(\mathcal{P}_k) = \exp\left(-\frac{1}{2}\|z_k - \mu_k\|_2^2\right). \quad (13)$$

A high α_k signifies that the instance prototype $\mathcal{P}_k$ satisfies its representative class distribution. The fusion therefore places more trust in the assembled visual attribute A_k^{vis} . A low α_k signifies that $\mathcal{P}_k$ is far from its representative class distribution. This indicates a low-quality or atypical sample. The fusion mechanism will then fall back on the stable, general text anchor T_k . The final fused prototype Q_k is a linear interpolation controlled by this score:

$$Q_k = (1 - \alpha_k(\mathcal{P}_k))T_k + \alpha_k(\mathcal{P}_k)A_k^{vis}(\mathcal{P}_k). \quad (14)$$

This Q_k replaces the static T_k as query prototype in the CAM calculation in Eq. (1) which provides a query prototype that dynamically balances text semantic generality with instance-specific visual descriptions. The CAMs are then refined to generate dense pseudo-labels $\hat{P}$ as the supervision of decoder prediction P .

3.4. Decoder Semantic Enhancement

To further enhance the semantic consistency of the decoder embeddings, we use a learnable adapter to transfer the frozen CLIP features to the decoder and introduce contrastive learning that aligns the adapted instance-specific features with the stable, global class prototypes from our inter-class H-GMM. Specifically, the adapter f_{adapt} maps the frozen CLIP features V to V_{dec} , and we can get the adapter instance prototype $\mathcal{P}_{dec,k}$ by mask average pooling as Eq. (2). The global semantic anchors G_k are derived from the component centers $\{\mu_k\}_{k=1}^K$ of inter-class GMM, mapping them back to the visual space $V_k^g = g_\theta(\mu_k)$. Then, we use an InfoNCE loss which pulls the adapter instance prototype $\mathcal{P}_{dec,k}$ towards its corresponding global anchor G_k while pushing it away from anchors of other classes:

$$\mathcal{L}_{con} = -\log \frac{\exp(\text{sim}(\mathcal{P}_{dec,k}, G_k)/\tau)}{\sum_{j=1}^K \exp(\text{sim}(\mathcal{P}_{dec,k}, G_j)/\tau)}, \quad (15)$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity and τ is a temperature. This process aligns the adapter representations with the global semantic relationship learned by our inter-class GMM. The enhanced semantic consistency adapter features V_{dec} are then fed into the decoder to generate the final segmentation prediction P .

3.5. Overall Framework Training

Our framework consists of two main separated training objectives. One is segmentation network training which uses

cross-entropy loss $\mathcal{L}_{ce} = \text{CE}(\hat{P}, P)$ and our contrast enhance loss $\mathcal{L}_{con}$. The total loss is:

$$\mathcal{L}_{seg} = \mathcal{L}_{ce} + \lambda \mathcal{L}_{con}, \quad (16)$$

where λ is the loss balance term. Another objective is INN training, which trains our probabilistic model using the loss $\mathcal{L}_{inn}$ in Eq. (9). In our training pipeline, the optimization of the INN and the segmentation network is completely independent, i.e., the gradients from the INN are not back-propagated to the segmentation network. And during inference process, the INN-related components are removed.

4. Experiments and Results

4.1. Experimental Settings

Datasets and Metrics. We evaluate our method on two standard WSSS datasets: PASCAL VOC 2012 [6] and MS COCO 2014 [16]. PASCAL VOC contains 21 classes and is extended with SBD dataset [8] following previous method [35, 46, 55]. The train, val, and test set consist of 10582, 1449, and 1456 images, respectively. MS COCO contains 81 classes, with train and val sets comprising 82,081 and 40,137 images, respectively. Mean Intersection-Over-Union (mIoU) is used as the main evaluation metric.

Implementation Details. We adopt ViT-B as the architecture for the frozen CLIP image encoder and enhance its fine-grained knowledge as in [42]. The architecture of segmentation decoder and the adapter also follow [42]. The architecture of INN mainly follow the RealNVP [4]. For hyperparameters, we set the reliable instance prototype filtering threshold η to 0.95, the number of visual attributes to 8 for each class, and the loss term λ in Eq. (16) as 0.2. For training on PASCAL VOC, batch size is 4, while for MS COCO, it is set to 8. The batch size of reliable prototype set $\mathcal{B}$ is 16. The adapter and decoder are trained using the AdamW optimizer with learning rate 1e-4 and weight decay 1e-2. The INN is trained separately using the Adam optimizer with learning rate 2e-4 and weight decay 1e-4. The total training iteration is set as 30,000 for VOC and 100,000 for COCO. All experiments are conducted on a single RTX 3090 GPU. Appendix for more details.

4.2. Comparisons with State-of-the-art Methods

Semantic Segmentation Results. Tab. 1 presents the semantic segmentation performance of our VDA and recent SOTA methods on the VOC and COCO datasets. To demonstrate the superiority of our method, we compare its performance with single-stage and multi-stage WSSS methods. Our single-stage VDA achieves 79.9% and 79.8% mIoU on the VOC val and test sets, respectively, as well as 51.4% mIoU on the COCO val set. It can be found that VDA outperforms all previous multi-stage methods across the board. Compared with the SOTA single-stage method ExCEL, it

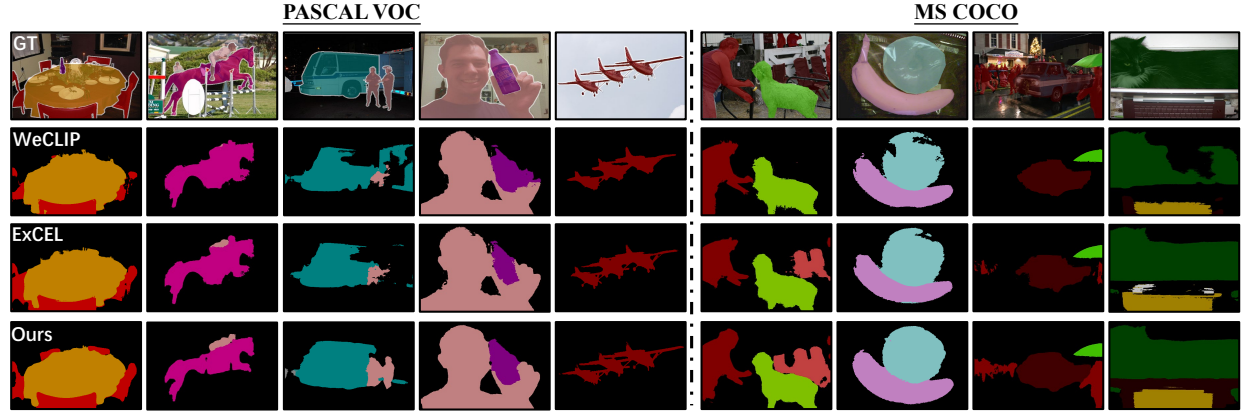


Figure 3. Segmentation visualizations of WeCLIP [48], ExCEL [42] and ours on PASCAL VOC and MS COCO datasets. Our method generates more precise segmentation results compared with other methods.

Table 1. Segmentation comparisons on VOC and COCO. Net. is the backbone for segmentation. Sup. is the supervision type. I: image-level labels. S: saliency maps. L: language.

Method	Net.	Sup.	VOC		COCO
			Val	Test	Val
<i>Multi-stage weakly supervised approaches</i>					
L2G _{CVPR'22} [11]	RN101	I+S	72.1	71.7	44.2
SIPE _{CVPR'22} [2]	RN38	I	68.8	69.7	43.6
MCTformer _{CVPR'22} [38]	RN38	I	70.4	70.0	42.0
USAGE _{CVPR'23} [20]	RN38	I	71.9	72.8	44.3
CLIMS _{CVPR'22} [37]	RN101	I+L	70.4	70.0	-
CLIP-ES _{CVPR'23} [17]	RN101	I+L	73.8	73.9	45.4
CPAL _{CVPR'24} [29]	RN101	I+L	74.5	74.7	46.8
S2C _{CVPR'24} [15]	SAM	I	78.2	77.5	49.8
VPL _{AAAI'25} [39]	ViT-B	I+L	79.3	79.0	49.8
POT _{CVPR'25} [33]	RN50	I+L	76.1	76.7	47.9
<i>Single-stage weakly supervised approaches</i>					
AFA _{CVPR'22} [26]	MiT-B1	I	66.0	66.3	38.9
ToCo _{CVPR'23} [27]	ViT-B	I	71.1	72.2	42.3
SeCo _{CVPR'24} [41]	ViT-B	I	74.0	73.8	46.7
DuPL _{CVPR'24} [36]	ViT-B	I	73.3	72.8	44.6
DIAL _{ECCV'24} [10]	ViT-B	I+L	71.4	71.4	44.4
WeCLIP _{CVPR'24} [48]	ViT-B	I+L	76.4	77.2	47.1
ExCEL _{CVPR'25} [42]	ViT-B	I+L	78.4	78.5	50.3
VDA(w/o CRF)	ViT-B	I+L	78.7	78.8	50.5
VDA(Ours)	ViT-B	I+L	79.9	79.8	51.4

shows a noticeable 1.5% improvement on VOC and 1.1% improvement on COCO. For the quantitative comparison results, we perform visualization and comparison between the segmentation masks of our VDA and those of WeCLIP and ExCEL in Fig. 3. By constructing a robust visual query prototype to replace the static text query, VDA achieves more precise object segmentation than SOTA methods that rely on the traditional image-text paradigm.

Localization Performance. Tab. 2 shows the quality of CAM seeds on VOC train set. It can be found that our VDA

Table 2. CAM seed comparisons on VOC train set. M: multi-stage methods. S: single-stage methods.

Method	Type	Net.	Sup.	VOC
				Train
LPCAM _{CVPR'23} [3]	M	RN50	I	65.3
MCTformer _{CVPR'22} [38]	M	RN38	I	61.7
AFA _{CVPR'22} [26]	S	MiT-B1	I	65.0
SFC _{AAAI'24} [51]	M	RN101	I	64.7
CLIMS _{CVPR'22} [37]	M	RN101	I+L	56.6
CLIP-ES _{CVPR'23} [17]	M	ViT-B	I+L	70.8
CPAL _{CVPR'24} [29]	M	RN101	I+L	71.9
S2C _{CVPR'24} [15]	M	SAM	I	76.1
VPL _{AAAI'25} [39]	M	ViT-B	I+L	77.8
DIAL _{ECCV'24} [10]	S	ViT-B	I+L	75.2
WeCLIP _{CVPR'24} [48]	S	ViT-B	I+L	75.4
ExCEL _{CVPR'25} [42]	S	ViT-B	I+L	78.0
VDA(Ours)	S	ViT-B	I+L	79.1

is significantly better than previous single-stage and multi-stage methods, achieves 79.1% mIoU. Furthermore, from the CAM visualization results in Fig. 4, it can be observed that compared with ExCEL, a previous SOTA CLIP-based WSSS method that only uses text as queries, our method fully leverages visual query information to achieve more complete and accurate CAMs.

4.3. Ablation Studies

Effect of the main components. To validate the efficacy of components in our framework, we conduct ablation studies in Tab. 3. Row I shows the performance which only uses text templates to generate CAM. In Row II, by incorporating VDAF (and its requisite Filter) into the baseline, we observe that the performance significantly jumps to 78.4% (a +2.6% gain), which confirms the importance of dynamic

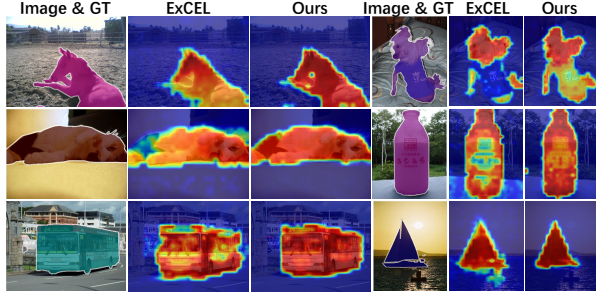


Figure 4. Visual comparison of CAMs on VOC train set. Compared with SOTA CLIP-based WSSS method ExCEL [42], our method achieves more complete and accurate activation.

Table 3. Ablation studies of the main components of our method on PASCAL VOC val set.

	VDAF	DSE	Filter	mIoU(%)
I				75.8
II	✓		✓	78.4
III		✓	✓	76.5
IV	✓	✓		75.3
V	✓	✓	✓	78.7

visual knowledge for generating better CAM supervision. Even using only DSE to enhance the decoder embeddings as shown in Row III, it improves the baseline performance to 76.5%. When both VDAF and DSE are enabled, our full model (Row V) achieves the peak performance of 78.7%, showing their synergistic benefits. Besides, Row IV shows that removing the Zero-shot Filter causes performance to catastrophically collapse from 78.7% to 75.3%. It proves that the Filter is an indispensable cornerstone of our method even if it is simple, as a high-quality latent space which both VDAF and DSE depend upon is critically reliant on a clean reliable batch of prototypes.

Effect of the Attributes Number. We investigate the impact of M , the number of visual attributes decoupled by our intra-class GMM in Tab. 4. It achieves best performance when M is set as a moderate number 8 which is enough to capture the intra-class attributes. When M is set to only 3 or 5, the small number of attributes fails to cover the varying intra-class attributes, leaving some attributes still entangled and ultimately resulting in lower performance. When M is increased further to 10 or 15, the performance begins to gradually decline, potentially due to the model learning overly fine-grained, less robust attribute representations. Moreover, a larger M increases the complexity of INN optimization, which is unnecessary.

Comparisons between Static Fusion and Our Dynamic Fusion. To validate our density-based dynamic fusion mechanism, we compare its performance against static fu-

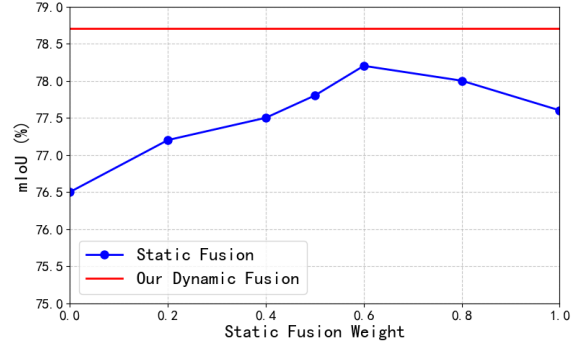


Figure 5. Sensitivity analysis of static fusion (between text prototype and visual prototype) and our dynamic fusion mechanism.

Table 4. Sensitivity analysis on VOC val set in terms of the number of attributes M within each class.

M	3	5	8	10	15
mIoU(%)	77.9	78.2	78.7	78.6	78.4

sion strategies. In this analysis, we replace our dynamic weight $\alpha_k(\mathcal{P}_k)$ in Eq. (14) with a fixed static weight α . As shown in Fig. 5, we vary α from 0 (text-only) to 1 (visual-only). The results show that both extremes are suboptimal, yielding 76.5% mIoU at $\alpha = 0$ and 77.6% mIoU at $\alpha = 1$. While using only the visual prototype still outperforms using only the text prototype, demonstrating the importance of the visual query information. The static fusion of these two is obviously beneficial, with the mIoU reaching its peak when $\alpha = 0.6$. Our dynamic fusion method achieves the best performance of 78.7% mIoU. It demonstrates that our density-based weighting, which adaptively evaluates the quality of each instance prototype, is more effective and robust than any fixed fusion weight.

5. Conclusion

In this paper, we tackled the inherent modality gap and static prototype limitations in CLIP-based WSSS. We introduced Visual Description Assembly (VDA), a novel framework that bypasses flawed text optimization by constructing dynamic, instance-specific prototypes directly from the visual space. We use learnable probabilistic modeling process to map entangled and scattered visual features to a structured latent space, explicitly disentangling and aggregating varied visual attributes. These attributes are then dynamically assembled based on the object response and adaptively fused with the stable text anchor, creating a robust query prototype to generate superior CAMs. Our approach achieves new SOTA performance on both PASCAL VOC 2012 and MS COCO 2014 benchmarks.

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